

Defining "Normal" EEG: The Gap Around Seizures

A plain-language note · 29 September 2026 · Kevin Kim

In short. To count false alarms fairly, we need stretches of brain activity that are clearly normal: far enough from any seizure that nothing seizure-related is going on. How far is "far enough" is a choice. We use 4 hours, a common choice in seizure-prediction research. A shorter gap gives more data, especially for the Siena patients, but risks counting post-seizure recovery or early pre-seizure changes as normal. Our decision: keep 4 hours as the main definition, and report a planned 1-hour analysis alongside it.

1. What the gap is

Every moment of a recording gets one of four labels: **before a seizure** (the 30 minutes in which a warning should come), **during a seizure**, **normal**, or **left out**. The gap decides where "normal" begins: only brain activity at least that far from any seizure, before or after, counts as normal. Everything between the gap and the seizure is left out.

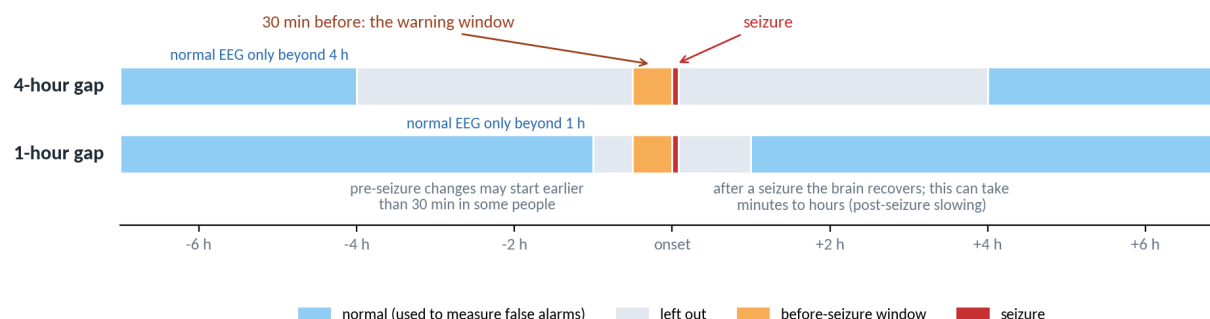


Figure 1. The same seizure labeled with a 4-hour gap (top) and a 1-hour gap (bottom). With 1 hour, much more of the recording counts as normal, but that normal EEG sits much closer to the seizure.

2. Why have a gap at all?

After a seizure, the brain needs time to recover. Brain waves are often slower and weaker for a while afterwards, for minutes and sometimes hours. That recovery isn't normal brain activity.

Before a seizure, changes may start early. Our warning window covers the last 30 minutes, but in some people changes may begin earlier. Brain activity just outside the window could still be "on the way" to a seizure.

If either kind of activity is labeled normal, the results are distorted. Post-seizure activity can look risky to the program, adding false alarms that aren't really false. Early pre-seizure changes labeled normal teach the program the wrong lesson. A generous gap keeps "normal" genuinely normal.

3. The options

Gap	Normal EEG available	Risk of mislabeling	Where it's used
1 hour	Most	Highest: some post-seizure recovery, and possibly early pre-seizure changes, may count as normal	Less conservative studies
2 hours	More	Moderate	A middle ground
4 hours	Least	Lowest	The Kaggle and Melbourne seizure-prediction competitions and many CHB-MIT studies (our choice)

Shorter than 1 hour isn't advisable. Normal EEG would then sit right next to the 30-minute warning window, and studies commonly set aside about the first hour after a seizure for recovery.

There is no single agreed standard. One review notes that researchers don't agree on these definitions, and that they are sometimes chosen after the fact to make results look better. That's why ours was fixed in the test plan before any result was seen.

4. What it means for our data

We counted, without training or testing anything, how much normal EEG and how many testable patients each gap would give:

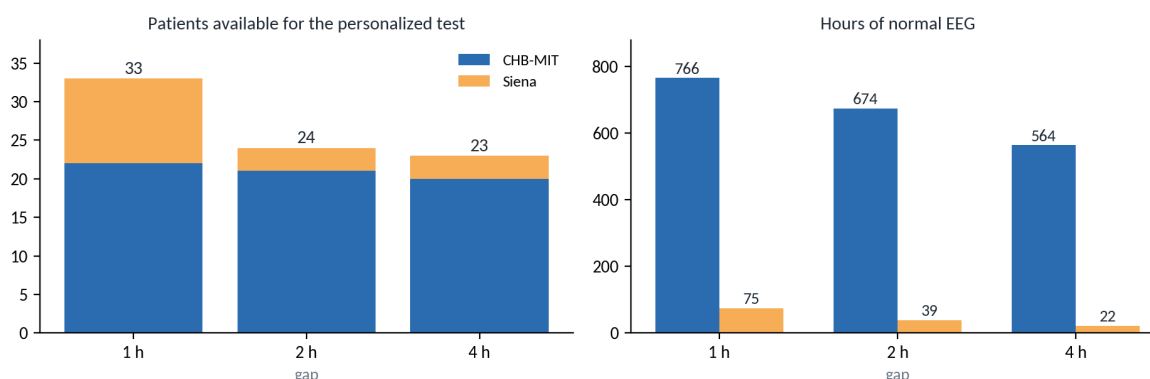


Figure 2. Left: patients with at least two usable seizures and an hour of normal EEG, who can take part in the personalized test. Right: hours of normal EEG.

Gap	CHB-MIT normal EEG	Siena normal EEG	Siena test patients	Personalized patients (CHB-MIT + Siena)
1 hour	766 h	75 h	13	22 + 11 = 33
2 hours	675 h	39 h	4	21 + 3 = 24
4 hours	564 h	22 h	4	20 + 3 = 23

2 hours adds almost nothing. 1 hour changes the picture for Siena: testable Siena patients go from 4 to 13, because Siena's seizures come close together, with most of the time between them lasting 1 to 4 hours. CHB-MIT recordings are long, so they already have plenty of normal EEG and barely change.

A small note: this count gives 23 personalized patients at 4 hours, while our runs used 22. The two methods measure normal EEG slightly differently (time ranges versus 30-second clips), so one patient sits right at the 1-hour minimum.

5. Our decision and plan

Main results keep the 4-hour gap. It's the most trustworthy definition of normal, it matches common practice, and it was fixed before testing. Everything reported so far uses it, and so will the final test on the sealed SeizureIT2 data.

A planned 1-hour analysis, reported alongside. After the final design is frozen, it will be tested once with a 1-hour gap on CHB-MIT and Siena, on about 33 patients instead of about 23, with more adults. If the conclusions hold under both definitions, they don't depend on this choice. If they differ, that's important to know too. Only the labels change for this, so the brain-wave measurements don't need recomputing.

An idea for later: train with 1 hour, test with 4 hours. The program could learn from the extra normal EEG a 1-hour gap provides, while still being tested only on truly normal EEG at least 4 hours from a seizure. That would add training data without making the test any easier. Whether it helps is an open question, to be planned and tested like every other change.

Sources

Brinkmann, B. H., et al. (2016). Crowdsourcing reproducible seizure forecasting in human and canine epilepsy. *Brain*, 139(6), 1713–1722 (the Kaggle competition).

Kaggle interictal definition, 4 hours from any seizure for humans: arxiv.org/pdf/2106.04510

Melbourne competition, at least 3 hours before and 4 hours after any seizure:

arxiv.org/pdf/2007.11716

A CHB-MIT study using 4 hours: arxiv.org/pdf/1707.01976

Excluding the hour after a seizure onset: arxiv.org/pdf/2110.13550

On the lack of agreed definitions: arxiv.org/pdf/2202.12598